

RoadSense

Not just where a speed limit is unsafe. Why, and what to do about it.

Findings Summary and Recommended Interventions

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Limit Check

ADB “AI for Safer Roads” Innovation Challenge

July 5, 2026

Abstract. Posted speed limits in Thailand and Maharashtra are frequently misaligned with what road conditions can safely support, but not always in the direction policymakers might expect. We built a Speed Safety Score (SSS) across 11,134 monitored road segments in Thailand and 3,577 in Maharashtra, grounded in the Safe System framework and a physics-based power-law risk model, that identifies not just where a posted limit is unsafe but why, so the recommended fix differs by segment. We flag 228 segments in Thailand (2.0%) and 137 in Maharashtra (3.8%) as statistically significant Speed-Unsafe Segments. In both countries the dominant recommended intervention is enforcement or traffic calming, not sign changes, an unintuitive finding with direct budget implications.

1. Introduction

The challenge asks whether posted speed limits are appropriate for their road context, not whether drivers currently obey them. The two are related but distinct: a segment can have a low rate of speeding while still posting a limit that is unsafe by Safe System standards, and vice versa. This distinction motivates our entire methodology and is the source of the paper’s central finding: a naive “misalignment” narrative implies limits are wrong and should be lowered, but our diagnostic shows this is the **minority** case in both countries.

2. Methodology

Our pipeline layers four signals into a single composite score, each targeting a distinct, named failure mode rather than a single opaque model.

Layer 1, Safe System Gap. Each segment’s `RoadClass` $\times$ `LandUse` combination maps to a Safe System survivable-speed threshold [1], e.g. 30 km/h for urban secondary roads with pedestrian mixing, 100 km/h for access-controlled motorways. Rather than a linear gap, we score risk as a power-law ratio

$$\text{SSG} = \left(\frac{v_{\text{posted}}}{v_{\text{threshold}}} \right)^n$$

following the Power Model of speed and crash severity [2], with $n = 4$ (fatal-crash exponent) for pedestrian-conflict cells and $n = 2$ (injury-crash exponent) elsewhere. Crash energy scales with v^2 and fatality risk climbs steeply with impact speed, so a road posted 20% over its safe threshold is meaningfully more dangerous than a linear model would suggest, not just 20% worse.

Layer 2, expected operating speed. A RandomForest regression predicts each segment’s 85th-percentile operating speed from road context alone (`RoadClass`, `LandUse`, length, traffic volume), deliberately excluding the posted limit, so the residual (actual minus predicted) is informative about whether drivers are responding to road design or ignoring it. Held-out R^2 is modest (0.39 Thailand, 0.43 Maharashtra) given the limited feature set available, we report this honestly rather than overstate precision. Every prediction uses genuine out-of-fold cross-validation, and each prediction’s reliability is weighted by the RandomForest’s own ensemble variance (agreement across its 300 trees) rather than a coarse sample-size

cutoff. This discount is what keeps a handful of data-thin road classes (e.g. Maharashtra’s 26-segment motorway class) from distorting the score.

Layer 3, peer comparison. Segments are grouped into peer cohorts (road class, land use, local population-density tier), and each segment’s posted limit is compared against its peers’ typical limit, surfacing outlier postings that a purely absolute threshold would miss. This layer’s peer cohorts also supply the reference statistics the significance test in Layer 4 uses, and the outlier finding itself is surfaced as a reported reason alongside a flagged segment, even though it is not summed into the numeric score directly, see below.

Layer 4, composite score and classification. The Safe System Gap and operating-speed residual combine into a single 0-100 Speed Safety Score via z-scored components, not independently percentile-ranked components, which we found compresses the top of the distribution into an indistinguishable plateau, modulated by a vulnerable-user-exposure (VUE) multiplier built from land use, traffic volume, and two open data layers: OpenStreetMap points of interest (schools, hospitals, markets) and WorldPop population density, sourced per the challenge FAQ’s own suggestion that `LandUse` alone is too coarse an exposure proxy. A segment is flagged as statistically significant when its score exceeds its Layer-3 peer cohort’s own mean by more than two standard deviations, with a minimum cohort-size floor so a small peer group’s noisy statistics cannot trigger a false flag. The continuous score and the significance flag combine into a four-tier classification, **Low**, **Moderate**, **High**, **Critical**, rather than leaving the audience to interpret a bare number: Critical and High require statistical significance (the misalignment is real, not peer-group noise), Critical additionally requires a score in the top decile.

Validation. No crash records exist for either country, so, following the same logic used in the surrogate-safety-measure literature [3], we benchmark the score against expected trends rather than a held-out ground truth: (1) internal consistency, flagged segments score far higher than unflagged ones (mean SSS 78-94 vs. 49-50), (2) concentration of the highest-risk segments in exactly the road categories Safe System doctrine predicts (urban secondary/pedestrian-conflict roads), (3) a weak-but-present correlation with independent speeding-behavior fields never used in scoring, deliberately weak, since a strong correlation would mean we were just re-measuring compliance rather than limit-appropriateness. Connected-vehicle research supports treating a purely kinematic, non-crash surrogate as a legitimate crash-risk indicator [4], reinforcing this approach.

3. Results

Metric	Thailand	Maharashtra
Segments analyzed	11,134	3,577
Flagged Speed-Unsafe (significant)	228 (2.0%)	137 (3.8%)
Median SSS, flagged segments	75.7 / 100	92.3 / 100
Mean stopping-distance excess, flagged	+52.5 m	+42.3 m
Recommended: lower the limit only	17 (7%)	26 (19%)
Recommended: road redesign	68 (30%)	45 (33%)
Recommended: enforcement / calming	142 (62%)	66 (48%)
Also high vulnerable-user exposure	53 (23%)	54 (39%)
Classification: Critical	35 (0.3%)	115 (3.2%)
Classification: High	193 (1.7%)	22 (0.6%)
Classification: Moderate	5,343 (48.0%)	1,652 (46.2%)
Classification: Low	5,563 (50.0%)	1,788 (50.0%)

Enforcement, not signage, is the majority fix in both countries. This is the single most policy-relevant finding. Our diagnostic shows that in 48-62% of flagged cases, the posted limit is broadly

acceptable but actual driving speed exceeds it, a behavior or road-design problem, not a signage problem. Conflating these would misdirect budget toward the wrong fix.

Maharashtra’s flagged segments are more concentrated and more severe (median flagged SSS 92.3 vs. 78.1, despite a smaller absolute count), consistent with sparser, lower-quality road infrastructure data producing fewer but starker outliers once the pipeline’s reliability gates (minimum peer-group size, ensemble-confidence weighting) are applied. This is the scalability story the challenge asks for: the same methodology, unmodified, degrades toward a smaller but higher-confidence flagged set in the data-sparse country, rather than failing or producing noise.

Vulnerable-user exposure compounds the risk for roughly a quarter to two-fifths of flagged segments. These should be treated as more urgent than their SSS alone indicates, since a misaligned limit near a school or market carries different stakes than the same misalignment on a low-pedestrian corridor.

4. Additional physics and statistical checks

Beyond the four-layer score, we implemented three further checks that produced working results rather than remaining theoretical.

Stopping-distance reinterpretation. Layer 2’s residual is re-expressed in physical units using the AASHTO stopping-sight-distance formula ($t_r = 2.5$ s perception-reaction time, $a = 3.4$ m/s² deceleration, both standard highway-design constants, not fitted to this data). The “stopping-distance excess” reported in Table 1 (Mean +52.5 m Thailand, +42.3 m Maharashtra for flagged segments) converts an abstract km/h residual into a concrete, policy-legible quantity: how many extra metres a driver needs to stop safely at the speed this road actually produces, versus what its design predicts.

Extreme Value Theory tail-risk modeling. We fit a Generalized Pareto Distribution to the upper tail of the composite risk index (peaks-over-threshold method, the same class of technique used for crash-probability tail estimation in the surrogate-safety literature) to estimate how extreme misalignment risk could get beyond the observed range. Thailand’s extrapolated and empirical 99.9th-percentile risk estimates agree within 1%, a good practical validation of the fit. The fit formally fails a Kolmogorov-Smirnov goodness-of-fit test in both countries, a known artifact of that test’s sensitivity at our sample sizes rather than evidence the tail model is wrong, so we report the result but do not lean on it as a rigorously validated statistical guarantee.

Street-level validation via Mapillary map features. For the statistically significant flagged segments (228 Thailand, 137 Maharashtra), we queried Mapillary’s map features API for nearby regulatory speed-limit signage and pedestrian-crossing infrastructure, a real, independently-sourced check against the fields already used in scoring. Coverage was far sparser in Maharashtra (12.4% of flagged segments had any nearby imagery) than Thailand (39.5%), consistent with the same data-sparsity pattern already documented in the network centrality and validation checks. Readable speed-limit signage was rare even where imagery existed, a genuine negative result we report rather than omit. The one segment where a sign was found is notable: a Maharashtra segment posted at 60 km/h has an actual street sign reading 30 km/h nearby, alongside 5 pedestrian-crossing features, a single concrete, independently-sourced instance of the exact posted-limit misalignment this project is built to detect. As a small demonstration of the vision-language-model path also scoped for this layer, we ran Qwen2-VL-2B on 13 sample street images (a rented GPU, not required for the map-features check above): sidewalks and marked crossings were visible in only 1 of 13, readable speed-limit signage in none, consistent with the map-features result. This is a proof of concept on a small sample, not a scored input, full details in the repo’s docs to detect.

5. Recommended interventions

- **Enforcement / traffic-calming (majority category):** speed cameras, physical traffic calming (speed humps, chicanes, narrowed lanes), or targeted police enforcement on segments where the limit itself looks reasonable but is routinely exceeded.

- **Road redesign:** segments where both the posted limit and actual driving speed exceed the Safe System threshold. A new sign will not fix this, the physical design (lane width, sightlines, lack of median separation) is inviting unsafe speeds regardless of what’s posted.
- **Lower the posted limit (sign-only):** the smallest category in both countries, segments where drivers are already driving close to a safe speed, so the posted limit is the only thing out of alignment.
- **Priority multiplier:** within any category, segments with high vulnerable-user exposure (schools, markets, dense population nearby) should be resequenced ahead of otherwise similarly-scored segments.

6. Limitations and future work

- **No crash ground truth exists** for either country, validation is necessarily trend-based, not outcome-based. We are explicit about this rather than implying a stronger empirical guarantee than the data supports.
- **LandUse is a coarse, GRUMP-derived urban/rural estimate** (per the challenge FAQ’s own caveat). We supplemented it with OpenStreetMap POI density and WorldPop population density (VUE tier 2), but a genuinely continuous exposure signal for Thailand (Maharashtra has one, UrbanPC) would strengthen this further.
- **Street-level visual features** (Mapillary imagery, vision-language model extraction of sidewalks, crossings, and road width) were scoped as a further enrichment tier but not implemented in this submission window. The SAGAI workflow [5] is a concrete, citable path to this, prioritizing binary/categorical visual prompts over continuous ones per that paper’s own accuracy findings.
- **Network-topology centrality was explored and found not usable:** the provided road segments form a simplified, sampled set of monitored links rather than a complete street network (72-83% of segments in our own constructed graph had zero betweenness due to fragmentation, varying by country), so this avenue was abandoned rather than reported as a false positive.

7. Conclusion and replicability

The pipeline is built entirely from fields present in the provided datasets plus two open, globally available enrichment sources (OpenStreetMap, WorldPop), with no country-specific tuning beyond the Safe System threshold table itself, which is derived from WHO/OECD doctrine rather than local calibration. Applying this methodology to a new country requires only: (1) the same TomTom-style probe/speed-limit dataset structure, (2) a RoadClass $\times$ LandUse Safe System threshold table, reusable as-is since it is policy-derived rather than data-fitted, and (3) an OpenStreetMap and WorldPop extract for the region, both free and global in coverage. The headline result, that enforcement rather than signage is the majority fix, is itself a replicable hypothesis other countries can test against their own data.

References

- [1] World Health Organization, “Speed Management: A Road Safety Manual for Decision-Makers and Practitioners,” technical report, 2008.
- [2] R. Elvik, P. Christensen, and A. Amundsen, “The Power Model of the Relationship Between Speed and Road Safety: Update and New Analyses,” *TOI Report*, no. 740/2004, 2004.
- [3] E. de Gelder *et al.*, “PRISMA: A Novel Approach for Deriving Probabilistic Surrogate Safety Measures for Risk Evaluation,” 2023.
- [4] Y. Li *et al.*, “From Lagging to Leading: Validating Hard Braking Events as High-Density Indicators of Segment Crash Risk,” 2026.
- [5] J. Perez and G. Fusco, “Streetscape Analysis with Generative AI (SAGAI): Vision-Language Assessment and Mapping of Urban Scenes,” 2025.